

BOMNAK, Russian Federation

WMO: 312530

Lat: 54.7118N

Lon: 128.8560E

Elev: 1196

StdP: 14.07

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-39.9	-36.4	-44.4	0.4	-39.1	-41.1	0.5	-36.0	15.2	13.1	12.7	7.3	0.3	0	0.619

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.5	83.1	66.7	79.7	64.9	76.5	63.4	69.1	78.7	67.1	76.4	65.4	73.8	2.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.9	100.0	72.9	63.9	92.9	71.3	62.0	86.7	70.0	34.1	78.9	32.4	76.4	31.0	73.9	80.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.3	13.5	11.3	-43.6	88.3	3.3	3.6	-45.9	90.9	-47.8	92.9	-49.7	94.9	-52.0	97.5		
(5)			-43.5	72.2	3.3	2.7	-45.8	74.1	-47.8	75.7	-49.6	77.2	-52.0	79.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	25.8	-18.8	-9.3	10.6	32.2	48.3	61.2	66.0	61.9	48.2	28.8	-1.3	-19.7		
	DBStd	32.36	10.46	10.11	10.96	8.77	7.14	6.57	5.72	5.86	7.45	9.67	15.06	11.10		
	HDD50	10158	2134	1661	1223	538	120	4	0	2	121	657	1538	2160		
	HDD65	14444	2599	2081	1688	985	519	147	56	130	505	1121	1988	2625		
	CDD50	1344	0	0	0	3	67	340	496	370	67	1	0	0		
	CDD65	156	0	0	0	0	1	33	87	34	1	0	0	0		
	CDH74	1265	0	0	0	0	23	323	668	246	5	0	0	0		
	CDH80	274	0	0	0	0	1	57	177	39	0	0	0	0		
	WSAvg	3.3	1.5	2.1	3.4	4.6	5.4	4.3	3.9	3.4	3.8	4.0	2.3	1.4		
	PrecAvg	22.5	0.2	0.2	0.4	1.3	2.0	3.1	4.9	4.3	3.0	1.5	0.8	0.4		
	PrecMax	30.9	0.7	1.2	1.6	4.0	5.4	7.0	13.3	11.5	6.3	4.6	2.2	1.1		
(15) Precipitation	PrecMin	14.4	0.0	0.0	0.0	0.1	0.4	0.8	0.4	0.3	0.7	0.2	0.2	0.0		
	PrecStd	4.3	0.2	0.2	0.4	0.9	1.1	1.5	2.5	2.2	1.5	1.0	0.5	0.3		
	DB	15.3	20.6	40.5	64.6	77.2	85.8	89.1	83.9	73.2	56.7	30.3	22.6			
	MCWB	13.0	16.8	32.2	48.5	57.2	67.2	68.5	68.2	61.4	46.5	26.3	18.2			
	DB	8.5	15.6	34.9	55.9	71.1	81.0	84.6	80.1	68.2	50.0	27.2	11.6			
(23) Mean Coincident Wet Bulb Temperatures	MCWB	6.4	12.7	28.3	43.1	54.2	63.8	67.5	65.9	57.9	41.7	25.1	9.3			
	DB	3.0	10.8	30.9	50.2	66.7	77.7	81.0	76.4	64.1	45.6	24.4	3.3			
	MCWB	1.7	8.6	25.6	39.8	51.8	62.0	66.5	64.3	54.8	39.2	22.1	2.1			
	DB	-2.7	6.7	27.2	45.6	62.4	74.4	77.4	72.9	60.4	41.9	20.4	-3.7			
	MCWB	-3.6	4.9	22.6	36.5	49.2	60.7	65.2	62.7	52.9	35.9	18.4	-4.5			
	WB	13.3	16.8	33.1	50.4	58.8	69.6	72.8	70.6	63.1	46.8	27.6	18.4			
	MCDB	15.2	20.3	40.2	64.1	73.8	79.2	81.8	80.9	71.2	55.6	29.0	22.3			
	WB	6.3	12.8	28.7	43.8	55.3	66.5	70.0	67.8	59.4	42.9	25.4	9.5			
	MCDB	8.3	15.5	34.0	54.9	67.9	77.6	79.5	76.4	66.2	48.5	27.3	11.5			
	WB	1.7	8.7	25.9	40.2	52.8	64.2	68.1	66.0	56.7	39.8	22.2	2.0			
(31) Mean Coincident Dry Bulb Temperatures	MCDB	3.0	10.6	30.7	49.4	64.7	74.3	77.9	73.6	61.9	45.1	24.4	3.2			
	WB	-3.6	4.9	22.7	37.0	50.3	62.1	66.4	64.0	54.3	36.6	18.4	-4.5			
	MCDB	-2.7	6.7	27.0	45.1	60.5	71.6	75.1	71.2	59.2	41.0	20.5	-3.8			
	MDBR	15.4	19.2	20.2	17.7	19.8	20.5	17.5	17.9	17.6	14.4	15.2	13.2			
	MCDBR	17.4	18.7	20.1	23.2	26.8	25.5	23.5	23.8	22.3	17.5	13.7	14.7			
(36) Mean Daily Temperature Range	MCWBR	15.9	16.4	15.4	14.1	14.0	11.8	10.6	11.5	13.3	11.6	11.8	13.5			
	MCDBR	17.4	18.3	19.2	22.2	24.6	22.0	20.1	20.0	18.5	16.5	13.2	14.2			
	MCWBR	15.9	16.0	15.0	14.0	13.6	11.7	10.7	10.9	11.9	11.3	11.8	13.0			
	taub	0.239	0.272	0.319	0.448	0.493	0.487	0.486	0.409	0.359	0.338	0.270	0.234			
	taud	2.143	2.113	2.039	1.860	1.923	2.042	2.085	2.310	2.385	2.228	2.119	2.085			
(42) Clear-Sky Solar Irradiance	Ebn at Noon	226	256	266	242	239	243	240	255	255	226	203	195			
	Edh at Noon	19	29	40	56	56	50	47	36	29	26	20	16			
	RadAvg	356	731	1297	1707	1656	1899	1652	1407	1020	649	378	256			
(45) All-Sky Solar Radiation	RadStd	14	28	54	115	115	130	172	112	89	55	24	10			

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	DBAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	DBAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page